An integral governed by a strip

the circle mean becomes the density $\mathrm{d}t/\cosh t$

$$\boxed{\int_0^\infty \frac{\sinh x}{x\cosh(x/a)}\,\mathrm{d}x = \log\cot\frac{\pi(1-a)}{4}, \qquad 0<a<1.}$$

Overview. The cosh in the denominator is not an accident of the integrand: it is the harmonic measure of the boundary of a horizontal strip, pushed forward from the uniform measure of a circle. The map

$$\Phi(\zeta) = \operatorname{Log}\frac{1+\zeta}{1-\zeta}$$

carries the unit disk onto the strip $|\operatorname{Im} w| < \pi/2$, and carries the uniform measure $\mathrm{d}\theta$ on the unit circle onto $\mathrm{d}t/\cosh t$ on the two boundary lines. The mean-value property for holomorphic functions therefore reads, in strip coordinates,

$$F(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{F(t+\frac{\pi\mathrm{i}}{2}) + F(t-\frac{\pi\mathrm{i}}{2})}{\cosh t}\,\mathrm{d}t,$$

and every integral in this note is one choice of F. Taking $F(w) = \mathrm{e}^{sw}$ gives $\int \mathrm{e}^{st}\operatorname{sech} t\,\mathrm{d}t = \pi\sec\frac{\pi s}{2}$; integrating that in the parameter gives the stated log cot.

0. The range of the parameter

The problem as posed does not specify which a are allowed, and the answer only makes sense for some. At the origin there is no difficulty:

$$\frac{\sinh x}{x\cosh(x/a)} \longrightarrow 1, \qquad x\to 0.$$

At infinity, for $a>0$,

$$\frac{\sinh x}{x\cosh(x/a)} \sim \frac{\mathrm{e}^{(1-1/a)x}}{x},$$

which is integrable exactly when $1-\frac{1}{a}<0$. Hence the working range is

$$\boxed{0<a<1,}$$

and note that the right-hand side $\log\cot\frac{\pi(1-a)}{4}$ is then finite and positive, blowing up as $a\to 1^-$ precisely where the integral diverges.

It is cleaner to prove a two-parameter statement and specialise.

Theorem. For real α,β with $\beta>0$ and $|\alpha|<\beta$,

$$\boxed{G(\alpha,\beta) = \int_0^\infty \frac{\sinh(\alpha x)}{x\cosh(\beta x)}\,\mathrm{d}x = \log\cot\frac{\pi(\beta-\alpha)}{4\beta}.} \qquad (*)$$

The integral of the problem is $G(1,1/a)$.

1. The disk becomes a strip

Let $\operatorname{Log} z = \log|z| + \mathrm{i}\operatorname{Arg} z$ with $-\pi < \operatorname{Arg} z < \pi$, and set

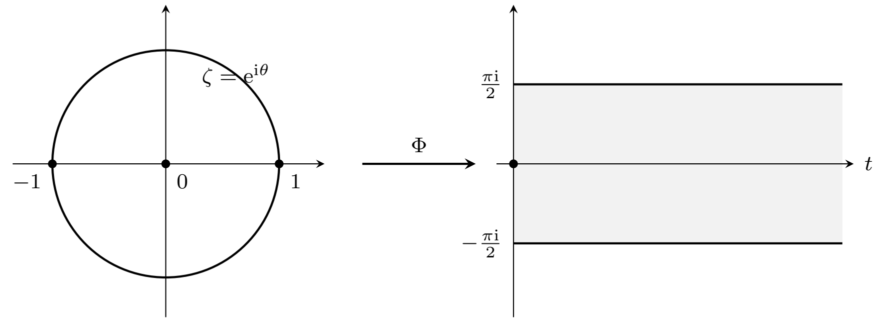
$$\Phi(\zeta) = \operatorname{Log}\frac{1+\zeta}{1-\zeta}, \qquad \zeta \in \mathbb{D} = \{|\zeta| < 1\}.$$

For $|\zeta| < 1$ the Möbius image $\frac{1+\zeta}{1-\zeta}$ lies in the right half-plane $\operatorname{Re} z > 0$: indeed

$$\operatorname{Re}\frac{1+\zeta}{1-\zeta} = \frac{1-|\zeta|^2}{|1-\zeta|^2} > 0.$$

So the principal branch is single-valued on $\mathbb{D}$, and since Log maps the right half-plane onto $\{|\operatorname{Im} w| < \pi/2\}$,

$$\Phi : \mathbb{D} \xrightarrow{\sim} S = \left\{w : \ |\operatorname{Im} w| < \frac{\pi}{2}\right\}, \qquad \Phi(0) = 0. \tag{1}$$



unit disk $\mathbb{D}$, uniform measure $\mathrm{d}\theta$ strip S, boundary measure $\mathrm{d}t/\cosh t$

2. The boundary measure is $\mathrm{d}t/\cosh t$

Put $\zeta = \mathrm{e}^{\mathrm{i}\theta}$. Factoring out the half-angle,

$$1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\theta/2}\cos\frac{\theta}{2}, \qquad 1 - \mathrm{e}^{\mathrm{i}\theta} = -2\mathrm{i}\,\mathrm{e}^{\mathrm{i}\theta/2}\sin\frac{\theta}{2},$$

so

$$\frac{1+\mathrm{e}^{\mathrm{i}\theta}}{1-\mathrm{e}^{\mathrm{i}\theta}} = \mathrm{i}\cot\frac{\theta}{2}. \tag{2}$$

For $0 < \theta < \pi$ we have $\cot\frac{\theta}{2} > 0$, so (2) is a positive multiple of i and

$$w = \Phi(\mathrm{e}^{\mathrm{i}\theta}) = t + \frac{\pi\mathrm{i}}{2}, \qquad t = \log\cot\frac{\theta}{2}. \tag{3}$$

For $\pi < \theta < 2\pi$ the cotangent is negative and the same computation gives $w = t - \frac{\pi\mathrm{i}}{2}$ with $t = \log\left|\cot\frac{\theta}{2}\right|$. The upper semicircle covers the upper boundary line, the lower semicircle the lower one; each covers it exactly once, since $\theta \mapsto \log\cot\frac{\theta}{2}$ is a decreasing bijection $(0,\pi) \to (+\infty, -\infty)$.

Now differentiate (3):

$$\frac{\mathrm{d}t}{\mathrm{d}\theta} = \frac{1}{\cot\frac{\theta}{2}} \cdot \left(-\tfrac{1}{2}\csc^2\tfrac{\theta}{2}\right) = -\frac{1}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = -\frac{1}{\sin\theta}. \tag{4}$$

Next, $\sin\theta$ can be expressed in t. From $\cot\frac{\theta}{2} = \mathrm{e}^{t}$, i.e. $\tan\frac{\theta}{2} = \mathrm{e}^{-t}$,

$$\sin\theta = \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} = \frac{2\mathrm{e}^{-t}}{1+\mathrm{e}^{-2t}} = \frac{1}{\cosh t}. \tag{5}$$

Combining (4) and (5),

$$\boxed{|\,\mathrm{d}\theta| = \frac{\mathrm{d}t}{\cosh t}.} \tag{6}$$

On the lower semicircle the sign of $\sin\theta$ flips but so does the orientation, and the same positive density results. This is the origin of the hyperbolic cosine: it is the Jacobian of the uniformising map, nothing more.

3. The mean-value property in the strip

For f holomorphic on $\mathbb{D}$ and continuous up to the boundary, the Cauchy formula at the centre reads

$$f(0) = \frac{1}{2\pi \mathrm{i}} \oint_{|\zeta|=1} \frac{f(\zeta)}{\zeta}\,\mathrm{d}\zeta = \frac{1}{2\pi}\int_0^{2\pi} f(\mathrm{e}^{\mathrm{i}\theta})\,\mathrm{d}\theta, \tag{7}$$

the average of f over the circle. Apply (7) to $f = F \circ \Phi$, which is holomorphic on $\mathbb{D}$ whenever F is holomorphic on S, and use $\Phi(0) = 0$ together with the change of variables (3), (6).

Mean-value property for the strip. Let F be holomorphic in $S = \{|\operatorname{Im} w| < \pi/2\}$, and set

$$G(\zeta) = F(\Phi(\zeta)), \qquad \zeta \in \mathbb{D}.$$

Assume that

(a) G extends continuously to $\overline{\mathbb{D}} \setminus \{1, -1\}$, the two boundary points that Φ sends to $w = \pm\infty$;

(b) the boundary values are absolutely integrable against the transported measure,

$$\int_{-\infty}^{\infty} \frac{\left|F(t + \frac{\pi \mathrm{i}}{2})\right| + \left|F(t - \frac{\pi \mathrm{i}}{2})\right|}{\cosh t}\,\mathrm{d}t < \infty; \tag{8}$$

(c) the two small arcs at $\zeta = \pm 1$ do not contribute:

$$\delta \sup_{\substack{|\zeta \mp 1| = \delta \\ |\zeta| \le 1}} |G(\zeta)| \longrightarrow 0, \qquad \delta \to 0^+. \tag{8$'$}$$

Then

$$\boxed{F(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{F\left(t + \frac{\pi \mathrm{i}}{2}\right) + F\left(t - \frac{\pi \mathrm{i}}{2}\right)}{\cosh t}\,\mathrm{d}t.} \tag{9}$$

Proof. For $0 < \delta < 1$ let

$$\mathbb{D}_\delta = \mathbb{D} \setminus \big(\{|\zeta - 1| \le \delta\} \cup \{|\zeta + 1| \le \delta\}\big).$$

Then $0 \in \mathbb{D}_\delta$, and by (a) the function G is holomorphic in $\mathbb{D}_\delta$ and continuous on its closure, so Cauchy's formula for this piecewise smooth Jordan domain applies at the centre:

$$G(0) = \frac{1}{2\pi \mathrm{i}} \int_{\partial \mathbb{D}_\delta} \frac{G(\zeta)}{\zeta}\,\mathrm{d}\zeta. \tag{9$'$}$$

The boundary $\partial\mathbb{D}_\delta$ consists of the unit circle with two short arcs removed, together with two arcs of radius δ about ± 1. On the latter, $|1/\zeta| \le (1-\delta)^{-1}$ and the arclength is $O(\delta)$, so their total contribution is $O\big(\delta \sup|G|\big) \to 0$ by (c). On the former, $\frac{\mathrm{d}\zeta}{\mathrm{i}\zeta} = \mathrm{d}\theta$, which by the change of variables (3), (6) becomes $\frac{\mathrm{d}t}{\cosh t}$ on each of the two boundary lines of S; the removed arcs correspond to $|t|$ large, and by the absolute convergence (8) the truncated integrals converge to the full ones. Letting $\delta \to 0^+$ in (9$'$) and using $G(0) = F(\Phi(0)) = F(0)$ gives (9). $\square$

The factor $1/\cosh t$ belongs to the change of variables, not to the mean-value theorem. Equivalently, $\frac{\mathrm{d}t}{2\pi \cosh t}$ on each of the two lines is the harmonic measure of the strip seen from the point $w = 0$.

4. Exponentials give the secant

Take

$$F(w) = \mathrm{e}^{sw}, \qquad s \in \mathbb{R}, \quad |s| < 1.$$

All three hypotheses hold, and checking them shows where the range $|s| < 1$ comes from.

(a) Continuity. Here

$$G(\zeta) = \left(\frac{1+\zeta}{1-\zeta}\right)^{s} = \exp\left(s \operatorname{Log} \frac{1+\zeta}{1-\zeta}\right),$$

and for $\zeta \in \overline{\mathbb{D}} \setminus \{\pm 1\}$ the Möbius image lies in the closed right half-plane minus the origin, where Log is continuous. So G extends continuously to $\overline{\mathbb{D}} \setminus \{\pm 1\}$.

(b) Integrability. On the boundary lines $|F| = \mathrm{e}^{st}$, and $\mathrm{e}^{st}/\cosh t$ decays like $\mathrm{e}^{-(1-|s|)|t|}$ at both ends, so (8) holds precisely when $|s| < 1$.

(c) The small arcs. Near $\zeta = 1$ one has $\frac{1+\zeta}{1-\zeta} \sim \frac{2}{1-\zeta}$, so on $|\zeta - 1| = \delta$

$$\delta\,|G| \asymp \delta \left(\frac{2}{\delta}\right)^{s} = 2^{s}\delta^{1-s} \longrightarrow 0 \quad \Longleftrightarrow \quad s < 1,$$

while near $\zeta = -1$, $\frac{1+\zeta}{1-\zeta} \sim \frac{1+\zeta}{2}$ gives $\delta\,|G| \asymp 2^{-s}\delta^{1+s} \to 0$ exactly when $s > -1$.

So the two endpoints of the strip impose the two halves of $|s| < 1$ separately: $s < 1$ comes from $w \to +\infty$ and $s > -1$ from $w \to -\infty$. The pole of $\sec \frac{\pi s}{2}$ at each end of the interval is the failure of the corresponding arc condition.

Since $F(0) = 1$ and

$$\mathrm{e}^{s(t+\pi\mathrm{i}/2)} + \mathrm{e}^{s(t-\pi\mathrm{i}/2)} = \mathrm{e}^{st}\left(\mathrm{e}^{\mathrm{i}\pi s/2} + \mathrm{e}^{-\mathrm{i}\pi s/2}\right) = 2\mathrm{e}^{st}\cos\frac{\pi s}{2},$$

formula (9) becomes

$$1 = \frac{\cos\frac{\pi s}{2}}{\pi}\int_{-\infty}^{\infty} \frac{\mathrm{e}^{st}}{\cosh t}\,\mathrm{d}t,$$

that is,

$$\boxed{\int_{-\infty}^{\infty} \frac{\mathrm{e}^{st}}{\cosh t}\,\mathrm{d}t = \pi \sec\frac{\pi s}{2}, \qquad |s| < 1.} \tag{10}$$

The pole of the secant at $s = \pm 1$ is exactly the divergence of the integral there, the same coincidence noted in Section 0, now visible as one identity.

Rescaling. Substituting $t = \beta x$ and $s = \alpha/\beta$ in (10), for $\beta > 0$ and $|\alpha| < \beta$,

$$\int_{-\infty}^{\infty} \frac{\mathrm{e}^{\alpha x}}{\cosh(\beta x)}\,\mathrm{d}x = \frac{\pi}{\beta}\sec\frac{\pi\alpha}{2\beta}. \tag{11}$$

The odd part of $\mathrm{e}^{\alpha x}$ integrates to zero over the symmetric line, so $\mathrm{e}^{\alpha x}$ may be replaced by $\cosh(\alpha x)$ at the cost of a factor 2:

$$\boxed{\int_{0}^{\infty} \frac{\cosh(\alpha x)}{\cosh(\beta x)}\,\mathrm{d}x = \frac{\pi}{2\beta}\sec\frac{\pi\alpha}{2\beta}.} \tag{12}$$

5. Integrating the parameter

Consider

$$G(\alpha, \beta) = \int_{0}^{\infty} \frac{\sinh(\alpha x)}{x\cosh(\beta x)}\,\mathrm{d}x, \qquad |\alpha| < \beta,\ \beta > 0.$$

Convergence is clear: near 0 the integrand tends to α, and at infinity $|\alpha| < \beta$ forces exponential decay. Differentiation under the integral sign is legitimate on any compact subinterval $|\alpha| \le \alpha_0 < \beta$, since

$$\left|\frac{\partial}{\partial\alpha}\frac{\sinh(\alpha x)}{x\cosh(\beta x)}\right| = \frac{\cosh(\alpha x)}{\cosh(\beta x)} \le \frac{\cosh(\alpha_0 x)}{\cosh(\beta x)},$$

and the majorant is integrable on $[0,\infty)$ by (12). Hence, by (12),

$$\frac{\partial G}{\partial\alpha} = \int_0^\infty \frac{\cosh(\alpha x)}{\cosh(\beta x)}\,\mathrm{d}x = \frac{\pi}{2\beta}\sec\frac{\pi\alpha}{2\beta}. \tag{13}$$

Since $G(0,\beta) = 0$, integrating (13) from 0 to α and substituting $v = \frac{\pi u}{2\beta}$,

$$G(\alpha,\beta) = \int_0^\alpha \frac{\pi}{2\beta}\sec\frac{\pi u}{2\beta}\,\mathrm{d}u = \int_0^{\pi\alpha/(2\beta)} \sec v\,\mathrm{d}v.$$

Because $|\alpha| < \beta$ forces $|v| < \pi/2$, the secant is positive throughout and no absolute value is needed:

$$\int \sec v\,\mathrm{d}v = \log(\sec v + \tan v),$$

so

$$G(\alpha,\beta) = \log\left(\sec\frac{\pi\alpha}{2\beta} + \tan\frac{\pi\alpha}{2\beta}\right).$$

Finally, the half-angle identity

$$\sec u + \tan u = \frac{1+\sin u}{\cos u} = \tan\left(\frac{\pi}{4} + \frac{u}{2}\right) = \cot\left(\frac{\pi}{4} - \frac{u}{2}\right)$$

with $u = \frac{\pi\alpha}{2\beta}$ turns this into

$$\boxed{G(\alpha,\beta) = \log\cot\frac{\pi(\beta-\alpha)}{4\beta},} \tag{14}$$

which is $(*)$.

6. The stated case

Take $\alpha = 1$, $\beta = \frac{1}{a}$. The hypothesis $|\alpha| < \beta$ is precisely $0 < a < 1$, matching Section 0. Then

$$\frac{\pi(\beta-\alpha)}{4\beta} = \frac{\pi\left(\frac{1}{a} - 1\right)}{4/a} = \frac{\pi(1-a)}{4},$$

so

$$\boxed{\int_0^\infty \frac{\sinh x}{x\cosh(x/a)}\,\mathrm{d}x = \log\cot\frac{\pi(1-a)}{4}, \qquad 0 < a < 1.}$$

A few checks. At $a = \frac{1}{2}$ the right side is $\log\cot\frac{\pi}{8} = \log(1+\sqrt{2}) = 0.8813\ldots$; numerical quadrature of $\int_0^\infty \frac{\sinh x}{x\cosh 2x}\,\mathrm{d}x$ agrees. As $a \to 0^+$, $\log\cot\frac{\pi(1-a)}{4} \to \log 1 = 0$, matching the fact that $\cosh(x/a)$ then crushes the integrand. As $a \to 1^-$ the right-hand side diverges like $\log\frac{4}{\pi(1-a)}$, the logarithmic divergence of $\int^\infty \frac{\mathrm{d}x}{x}$ predicted in Section 0.

7. What the argument shows

Residues would also evaluate the integral, but they would not explain the shape of the answer. The chain here is

$$\boxed{\mathbb{D} \longrightarrow \{|\operatorname{Im} w| < \tfrac{\pi}{2}\} \longrightarrow \frac{\mathrm{d}t}{\cosh t} \longrightarrow \pi \sec \frac{\pi s}{2} \longrightarrow \log \cot \frac{\pi(1-a)}{4}}$$

and each arrow is forced. The cosh in the denominator of the original problem is the harmonic-measure density of the strip; the secant is the mean of an exponential against that density; and the logarithm of a cotangent is the antiderivative of a secant. The appearance of cot in the answer to a problem stated purely with sinh and cosh is therefore not a coincidence of tables. It is the inverse of the very substitution $t = \log \cot \frac{\theta}{2}$ that produced the strip in Section 2.

Other choices of F in (9) give further identities with no extra work. For instance $F(w) = w^{2n}$ is holomorphic in S but fails (8); $F(w) = \mathrm{e}^{sw} w$ gives, on differentiating (10) in s,

$$\int_{-\infty}^{\infty} \frac{t\,\mathrm{e}^{st}}{\cosh t}\,\mathrm{d}t = \frac{\pi^2}{2} \sec \frac{\pi s}{2} \tan \frac{\pi s}{2},$$

and setting $s = 0$ in successive derivatives recovers the classical moments $\int_{-\infty}^{\infty} t^{2n} \operatorname{sech} t \,\mathrm{d}t$.